GAO YUNZHE

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EDUCATION

Nanyang Technological University (NTU) Singapore
M.Sc. in Computer Control and Automation **Aug. 2025 – Dec. 2026 (Expected)**

- **GPA: 4.63 / 5.0**
- Honors: EEE Global Merit Award (Top 2% of the cohort; ∼20 recipients out of 1,000+ eligible students).
- Advisor: Prof. Mao Kezhi
- Dissertation Topic: Design and Deployment of Lightweight Large Language Models (LLM) for Efficient, Localized Deployment.
- Selected Coursework: Artificial Intelligence & Data Mining (A); Genetic Algorithms & Machine Learning (A); Computer Control Systems (A); Robotics & Intelligent Sensors (A+)

Northeastern University (NEU) Shenyang, China
B.E. in Automation **Sept. 2021 – Jul. 2025**

- **GPA: 3.76 / 5.0 (Avg. Score: 87.6/100)**
- Honors: University Academic Scholarship.
- Selected Coursework: Distributed Cooperative Control and Optimization of Multi-Agent Systems (96/100); Artificial Intelligence Foundation (93/100); Introduction to Intelligent Control (97/100)

RESEARCH EXPERIENCE

Energy-Efficient Edge-Cloud Collaborative LLM Inference NTU, Singapore
Master's Dissertation (Advisor: Prof. Mao Kezhi) Aug. 2025 – Present

- Proposed "Green-Edge-Cloud," a router-free collaborative inference framework that uses the edge model's length-normalized sequence likelihood as an intrinsic confidence signal for query offloading.
- Implemented routing-oriented chain-of-thought distillation from Qwen2.5-72B-Instruct to Qwen2.5-1.5B-Instruct to improve confidence–correctness alignment for edge-cloud routing.
- Built an inference and energy-profiling pipeline using vLLM and CodeCarbon, accounting for edge idle energy during cloud offloading.
- Reduced total inference energy consumption by 81.16% and achieved a 5.6× inference speedup relative to cloud-only inference in system evaluations.

Democratizing Fine-Grained Manipulation via Imitation Learning NEU, China
Undergraduate Thesis (Advisor: Prof. Li Zhi) Nov. 2024 – Jun. 2025

- Built and deployed a 6-DOF Koch v1.1 leader-follower system for ACT-based imitation learning, integrating motor calibration, demonstration collection, policy training, and real-time control.
- Optimized task-specific inference pipelines: MobileNetV2 reduced plug-task latency from 20 to 16 ms and supported 60 Hz control, while YOLOv5 segmentation with asynchronous inference achieved 14–15 ms latency for conveyor sorting.
- Evaluated 50 plug-insertion trials and 750 conveyor-sorting operations, achieving 86% plug-grasp success, 66% insertion success, and 89.3% overall sorting success.

Prediction of Ash Content in Coal via Machine Learning NEU, China
Undergraduate Researcher Jan. 2024 – Aug. 2024

- Contributed to a research project on predicting ash content in coal using near-infrared spectral data and PSO-ELM.
- Participated in model development, experimental evaluation, and manuscript preparation; the study achieved $R^2 = 0.98$ and RMSE = 0.79.
- First author of the resulting paper published in Control Engineering of China (Core Journal).

Digital Twin-based Mineral Processing Intelligent Decision System NEU, China
Core Algorithm Engineer Jan. 2023 – Nov. 2023

- Developed ECAS, a correlation-based retrieval module that matches equipment states against historical abnormal operating patterns to support monitoring and decision-making in a mineral-processing digital-twin platform.
- Co-authored ECAS V1.0, registered as Chinese computer software copyright with second-author credit.
- The overall project received the National Silver Award at the China International College Students' Innovation Competition and the Grand Prize at the NEU Innovation and Entrepreneurship Competition.

PUBLICATIONS

[1] **Y. Gao**, Z. Feng, K. Mao. "Green-Edge-Cloud: Routing without Routers via Confidence Calibration." *ICML 2026 Workshop (EIML)*. **Accepted**.

[2] **Y. Gao**, J. Zou, D. Xiao. "Prediction of Coal Ash Composition Based on Near-Infrared Visible Spectroscopy and PSO-ELM." *Control Engineering of China*, 2024. (**Core Journal**)

INTELLECTUAL PROPERTY

[1] **Mineral Processing Equipment Abnormal Pattern Retrieval and Index Correlation Analysis System (ECAS) V1.0**
K. Zhang, **Y. Gao** (2nd Author). *Computer Software Copyright (China)*, Reg. No. 2023SR1469715, Nov. 2023

[2] **Intelligent Venue Resource Allocation and Security Operation Visual Analysis Software (IVVAS) V1.0**
K. Zhang, P. Zhu, **Y. Gao** (3rd Author). *Computer Software Copyright (China)*, Reg. No. 2023SR1421636, Nov. 2023

SELECTED AWARDS

- **Global Merit Award**, Nanyang Technological University 2025-2026
- **National Silver Award (Top 1% nationwide)** Dec. 2023
  China International College Students' Innovation Competition
- **University Academic Scholarship**, Northeastern University 2024-2025
- **University Academic Scholarship**, Northeastern University 2021-2022

TECHNICAL SKILLS

- **Programming:** Python, C/C++, MATLAB, SQL.
- **Machine Learning & LLMs:** PyTorch, Transformers, PEFT (LoRA), TRL, scikit-learn, CoT Distillation, Confidence-Based Routing.
- **Robotics & Vision:** Imitation Learning, Action Chunking Transformer (ACT), ROS, MobileNetV2, YOLOv5, MuJoCo, Gazebo.
- **Inference & Profiling:** vLLM, Quantized Inference, Edge-Cloud Inference, CodeCarbon, Energy and Latency Profiling.
- **Systems & Tools:** Linux, Docker, CUDA, Git, LaTeX.